Ozan Linus Zephyr Kiratli, PhD

Bioinformatics Scientist

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Professional Summary

Bioinformatics scientist with 8+ years of computational biology experience and 3+ years developing and deploying AI/ML systems for federal research and public health applications. Proven ability to lead interagency projects, design novel ML evaluation frameworks, and distill complex analytical findings into compelling visualizations and presentations for diverse stakeholders. U.S. Citizen with active Tier 1 DoD security clearance.

Experience

Bioinformatician - Postdoctoral Fellow — Oct 2022–present

Walter Reed Army Institute of Research (WRAIR), *Silver Spring, MD* — 40 hr/wk

- Co-Principal Investigator on $500K interagency AI/ML project funded by DoD Biosurveillance Program. Evaluating and adapting protein structure prediction models (SimpleFold, Foldseek) for automated novel pathogen detection from global genomic surveillance data; developing evaluation metrics and validation frameworks to assess model performance and reliability; deploying approved models into DoD biosurveillance systems.
- Designed Random Forest model and novel evaluation metrics for predicting antigenic evolution in dengue viruses from genomic sequences to enable epidemiological forecasting and vaccine effectiveness assessment. Devised pairwise comparison strategy yielding 80,000 training samples from 400 viral genomes; applied supervised and unsupervised machine learning methods. Developed full-stack pipeline (R, Bash, C++) for data curation and model training, evaluation, and validation.
- Analyzing genomic surveillance data from chikungunya virus outbreak in Kenya (2025–present) to provide actionable public health insights for emerging infectious disease response. Applying advanced Bayesian phylogenetic methods (BEAST) to detect outbreak source and transmission dynamics; coordinating with WRAIR-Africa and GEIS leadership on surveillance activities.
- Coordinated interagency and international biosurveillance collaboration with GEIS Network, WRAIR-Africa, and Naval partners; delivered briefings and reports on AI/ML integration and program recommendations to diverse federal, military, and academic stakeholders including senior leadership and funding agency representatives.
- Led bioinformatics team (5 members) during team leader's 2-month absence. Coordinated cross-functional operations with other WRAIR-VDP teams, maintained project timelines, and provided regular updates to leadership and stakeholders.
- Participated in REDI-NET (Remote Emerging Disease Intelligence Network) biosurveillance system training with government, Navy, and academic partners; briefed WRAIR-VDP leadership on system capabilities and implementation considerations; at the request of GEIS leadership, served on advisory team for integration of REDI-NET pipeline into DoD networks and systems.
- At the request of department leadership, contributed to policy analysis on AI/ML implementation for troop health research and biosurveillance activities. Assessed scientific validity, operational benefits, and risk mitigation strategies for AI adoption; delivered analysis to inform leadership decisions on AI integration into research and surveillance operations.

Research and Teaching Assistant — Aug 2016–Aug 2022

University of Pennsylvania, *Philadelphia, PA*

- Independently developed expertise in computational biology and built custom bioinformatics pipeline for allele frequency analysis from pooled Next-Generation Sequencing samples; designed automated data processing workflows using Bash scripting and R in Linux environments.
- Designed and managed multiple independent research projects involving experimental design, large-scale data collection, statistical analysis, and interpretation; designed and built custom large-scale experimental setups and assays for data collection, including custom instrumentation using Arduino-based hardware. Developed custom analytical pipelines and computational tools for genomic data analysis from pooled Next-Generation Sequencing samples; applied advanced analytical methods including population genetics approaches to address contemporary research questions in evolutionary biology.
- Supervised and mentored 7 undergraduate research assistants. Provided training on scientific methods, experimental design, and computational analysis techniques.

Education

PhD in Biology, University of Pennsylvania, *Philadelphia, PA* — 2022

Concentration: Evolutionary Biology; **Certification:** College and University Teaching

MS in Biology, Middle East Technical University, *Ankara, Turkey* — 2016

Concentration: Evolutionary Biology

BS in Biology, Middle East Technical University, *Ankara, Turkey* — 2013

Minor: Physics

Skills

Statistics & Evaluation: Advanced statistical inference (parametric, non-parametric, Bayesian); linear/non-linear regression; multivariate analysis (multivariate GLM, MANOVA, MANCOVA); uncertainty quantification; model evaluation frameworks; validation protocols; performance benchmarking; bias detection.

AI/ML Methods: Machine learning (Random Forest, supervised/unsupervised); deep learning (TensorFlow, Keras); protein structure prediction (SimpleFold, Foldseek); predictive modeling.

Software Development: R, Python, Bash, C++ , AWK, SQL; Nextflow, Docker, Conda, Git.

Computing Infrastructure: HPC/cloud computing, Linux/UNIX (Bash/Zsh), SLURM, distributed computing, containerization.

Bioinformatics: *Sequence alignment:* BWA, Bowtie2; *Variant calling and filtering:* Samtools, Bcftools, VCFtools; *Genomic data wrangling:* seqtk, IGV; *Phylogenetics:* IQ-TREE, BEAST; *Custom tool and pipeline development:* PoolSeqFlow (full-stack Illumina-to-allele-frequency pipeline), poolseq-VCF2Freq, FilteringFalsePositives, NGS-pipeline.

Visualization: ggplot2, Plotly, Jupyter notebooks, Quarto, LaTeX, MS Office Suite.

Honors and Awards

Co-Principal Investigator — Present
Project: AI/ML Deployment for Biosurveillance (USAMRIID, WRAIR, and NMRC collaboration) — Awarded: ~$500,000
DoD Biosurveillance Program (BSVp) – through GEIS, *Silver Spring, MD*

Postdoctoral Fellowship Recipient — 2025–Present
Oak Ridge Institute of Science and Education (ORISE), *Silver Spring, MD*

Postdoctoral Fellowship Recipient — 2022–2025
Army Educational Outreach Program (AEOP), *Silver Spring, MD*

Honorable Mention for Best Student Talk — 2020
American Naturalist 2020, *Pacific Grove, CA*

Selected Publications

Kiratli, **O. L. Z.**, Wang, L., Huang, A. T., Cummings, D. A. T., Salje, H., et al. (n.d.). "Antigenic distance prediction of dengue viruses using random forest models". [In Preparation].

Kiratli, **O. L. Z.** (n.d.). *PoolSeqFlow: A Nextflow pipeline for allele frequency analysis from pooled Illumina sequencing data*. URL: `https://github.com/ozankiratli/PoolSeqFlow`. [In Preparation].

Mayekar, H., **Kiratli**, **O. L. Z.**, Chakraborty, A., Arya, H., Solanki, P., et al. (n.d.). "Seasonal Allelic Shifts Reveal Constrained Thermal Adaptation in Highland *Drosophila melanogaster*". *Heredity*. [Under Review].

Rudman, S. M., Greenblum, S., Hughes, R. C., Rajpurohit, S., **Kiratli**, **O.**, et al. (2019). "Microbiome composition shapes rapid genomic adaptation of *Drosophila melanogaster*". *Proceedings of the National Academy of Sciences of the United States of America* 116.40, pp. 20025–20032. ISSN: 0027-8424. DOI: `10.1073/pnas.1907787116`. URL: `https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6778213/`. [150+ citations].

Selected Presentations

Near Real Time Bioinformatics Reach Back Support via Digital Biobank — Sep 2025
GEIS NGS Summit 2025. *Silver Spring, MD*. (Talk)

Machine learning can reveal genomic signals associated with antigenic distance in dengue viruses — Nov 2024
American Society of Tropical Medicine and Hygiene (ASTMH) Annual Meeting 2024. *New Orleans, LA*. (Talk)

Effects of migration on rapid adaptation and genetics of dispersal in natural *D. melanogaster* populations — Jan 2020
American Naturalist 2020. *Pacific Grove, CA*. (Talk)

Outreach, Leadership and Leisure Activities

Organizer, Virtual Graduate Research in Ecology, Behavior, and Evolution (GREBE) 2021, *Online* — 2019–2021

- Led group of organizers for annual conference (with history of over 30 years) for graduate students from Penn, Princeton, Rutgers, Columbia, and Yale.
- Prepared, hosted, and broadcasted prerecorded talks; moderated live Q&A sessions on Zoom.

Small Electronics and Computing Projects with Raspberry Pi and Arduino — 2017–present

- Deployed home server infrastructure including VPN, DNS, cloud storage, and network routing using Raspberry Pi and Arduino.
- Developed custom hardware tools for laboratory applications, including dual water thermometer, and microscope webcam.

Bass Guitar Player — 2003–present

- Played as part of 7 bands with over 50 group performances, including nationally televised event.